

phase_topcap | 419 training pts

Velocity RMSE (absolute | relative to truth-RMS):

whole domain : 1.005 (rel 0.1266) [26695 cells]

Pressure RMSE : 13.85

Momentum-integral force:

|F| : 3.033e+05 [33167 surface pts]

F vector : [1.551e+05, 2.118e+05, 1.52e+05]

F true : [1.554e+05, 2.086e+05, 7.259e+04] |F_true|=270114

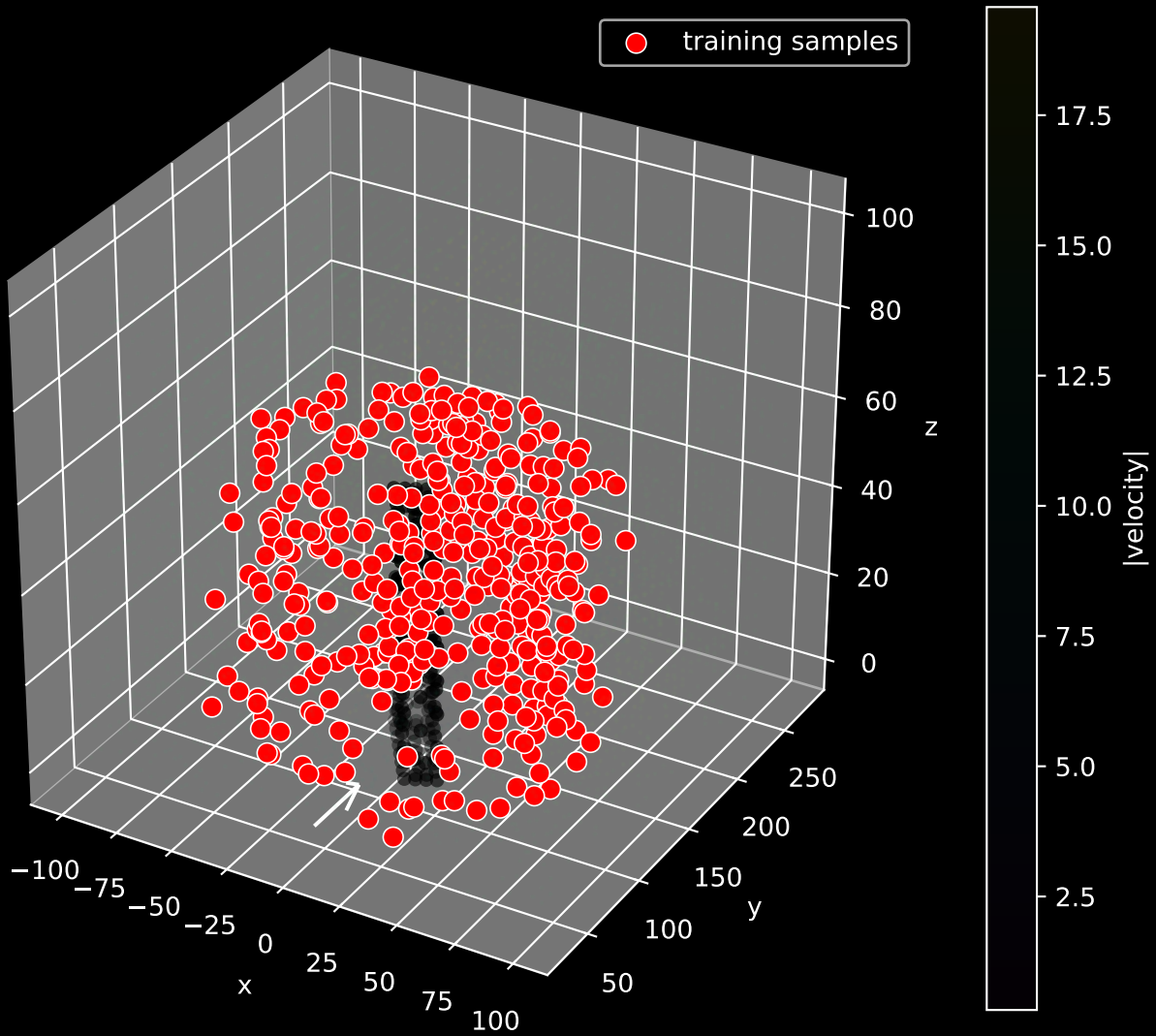
force error dF : [-373.4, 3110, 7.941e+04] rel=0.2942

Prior (potential-flow baseline) for reference:

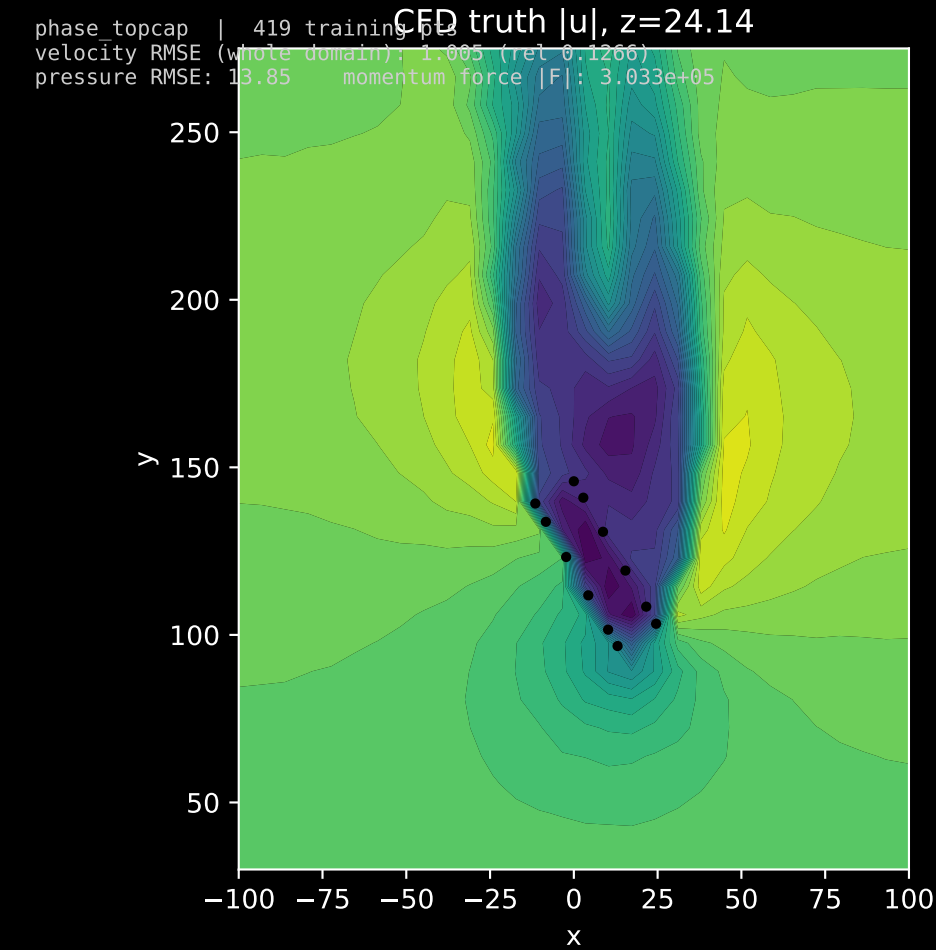
whole domain prior : 2.226

phase_topcap | 419 training pts
velocity RMSE (whole domain): 1.005 (rel 0.1266)
pressure RMSE: 13.85 momentum force |F|: 3.033e+05

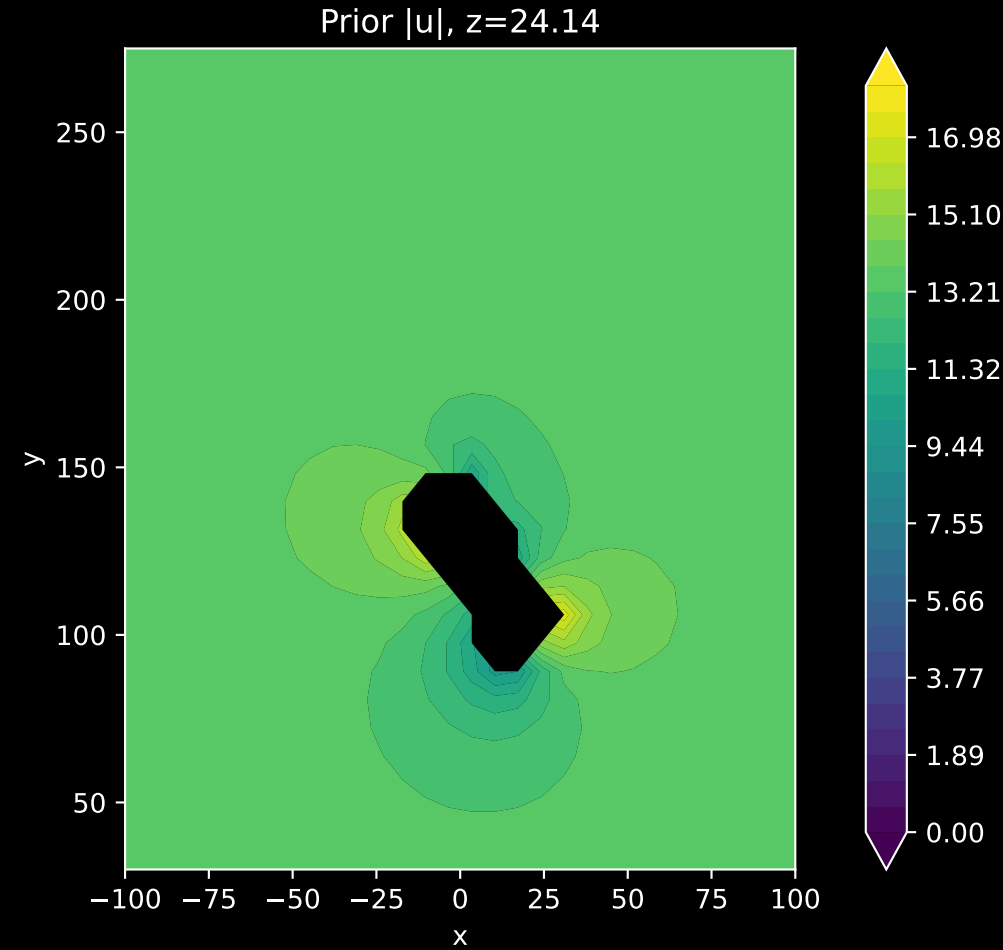
GPR posterior velocity magnitude



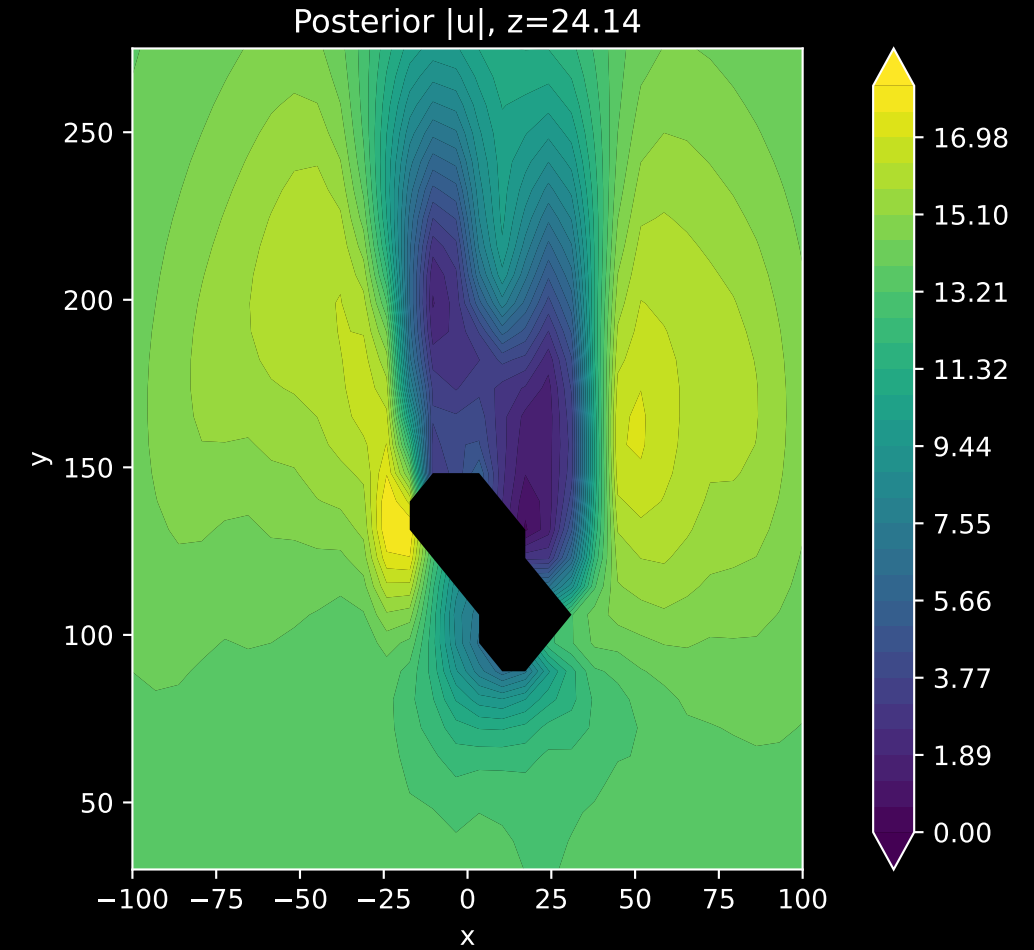
CFD truth $|u|$, $z=24.14$



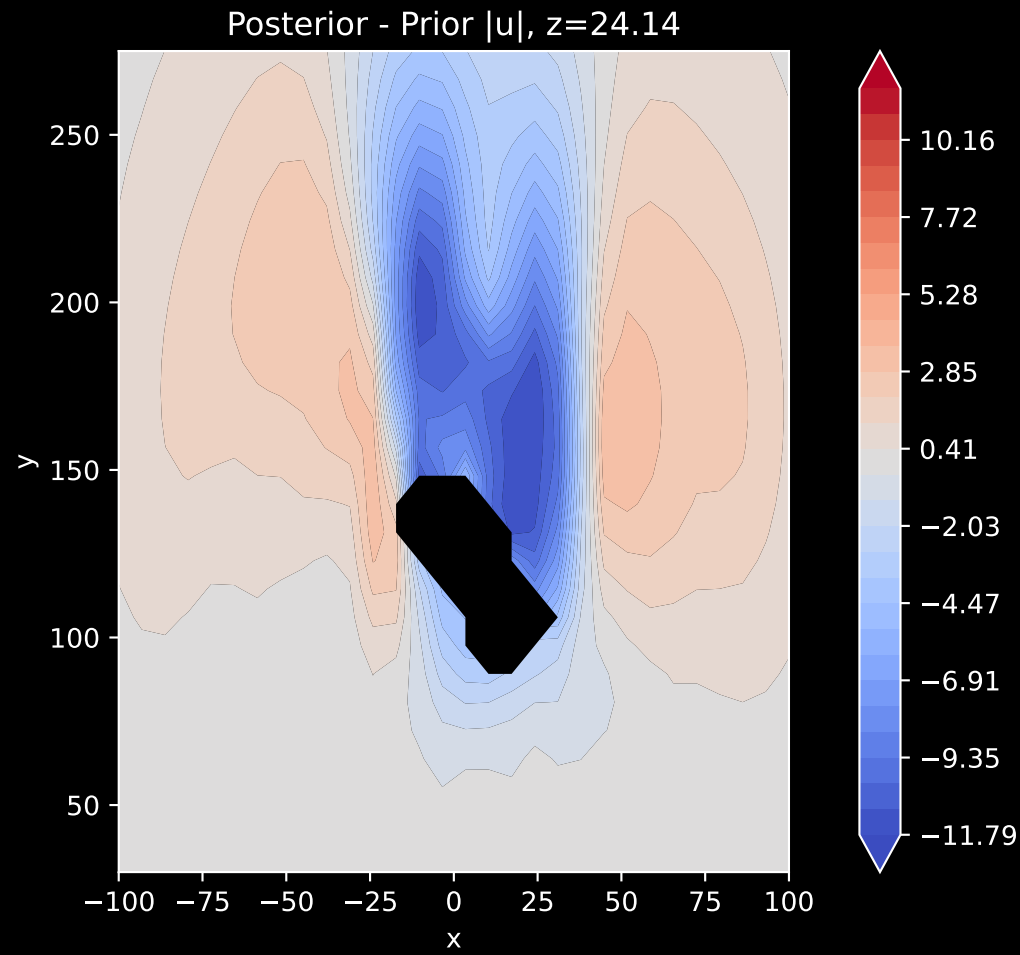
Prior $|u|$, $z=24.14$



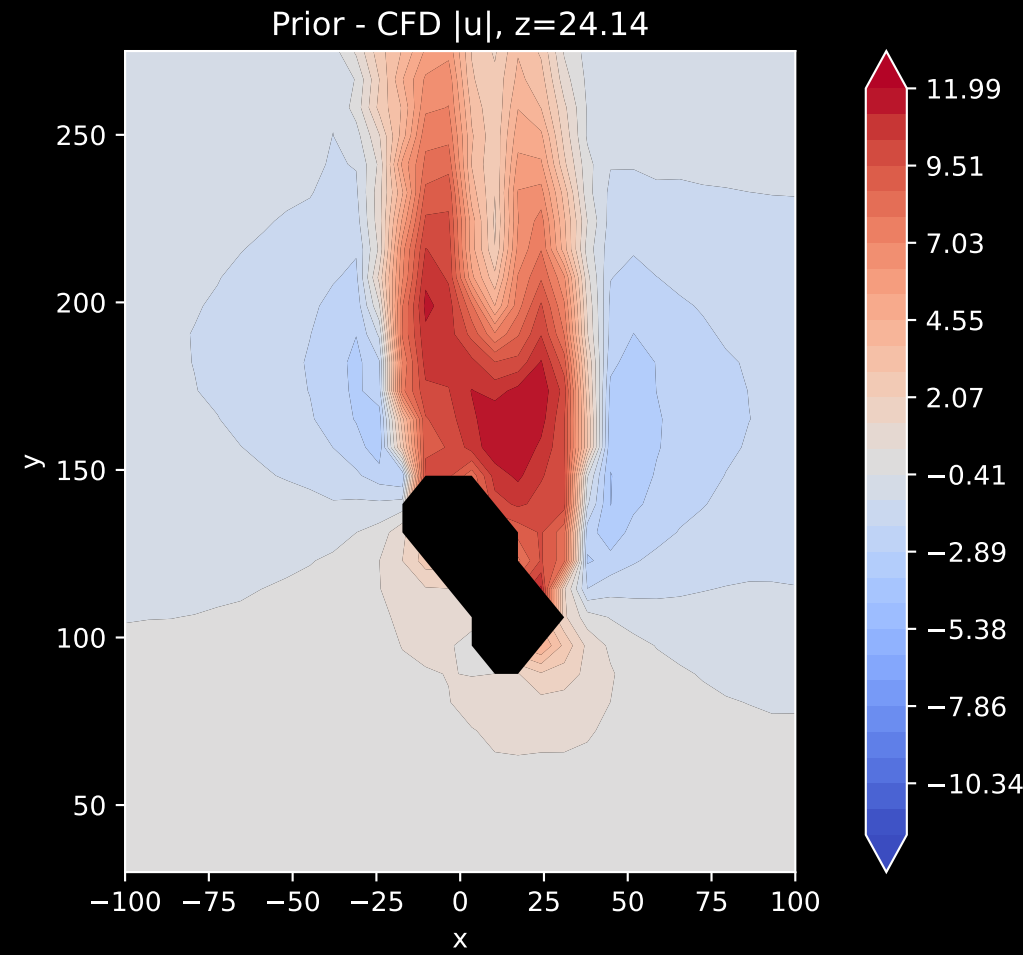
Posterior $|u|$, $z=24.14$



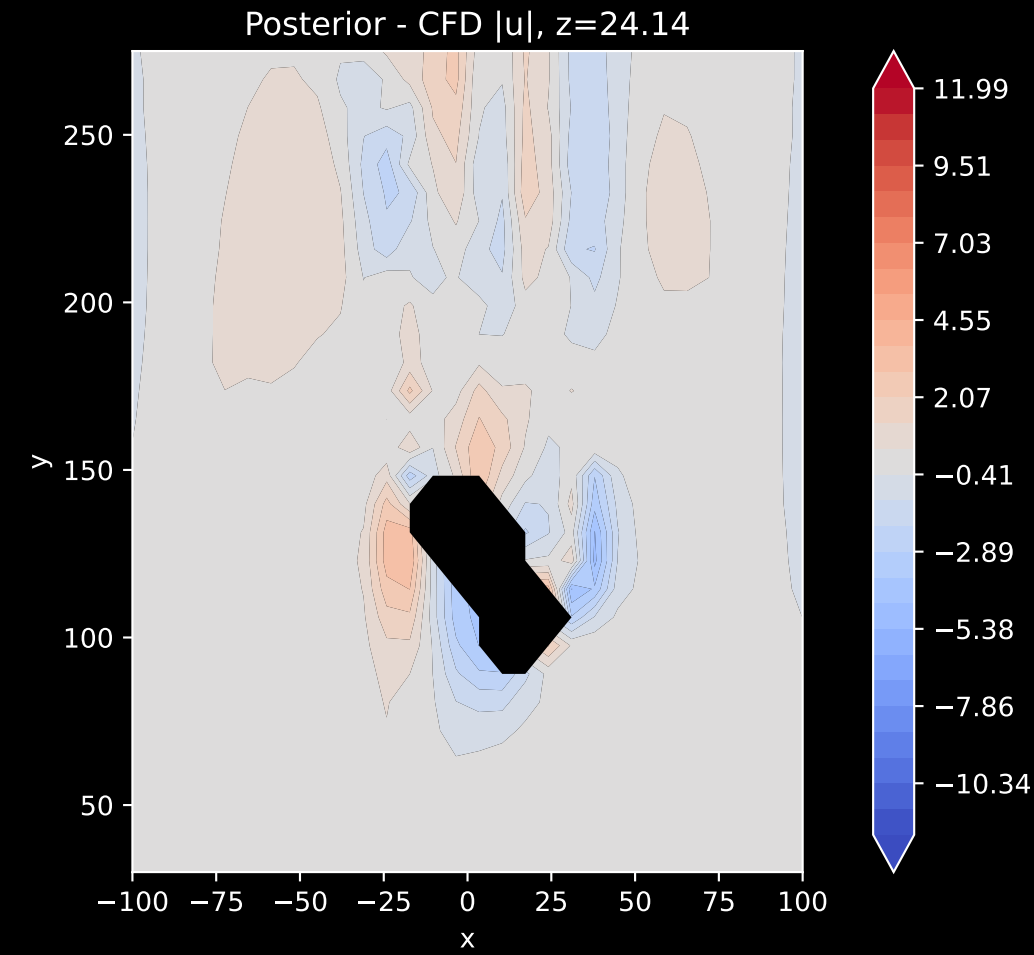
Posterior - Prior $|u|$, $z=24.14$



Prior - CFD $|u|$, $z=24.14$

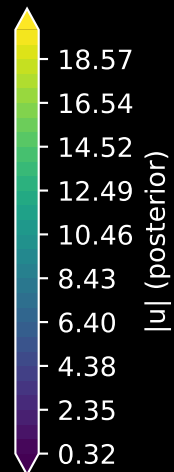
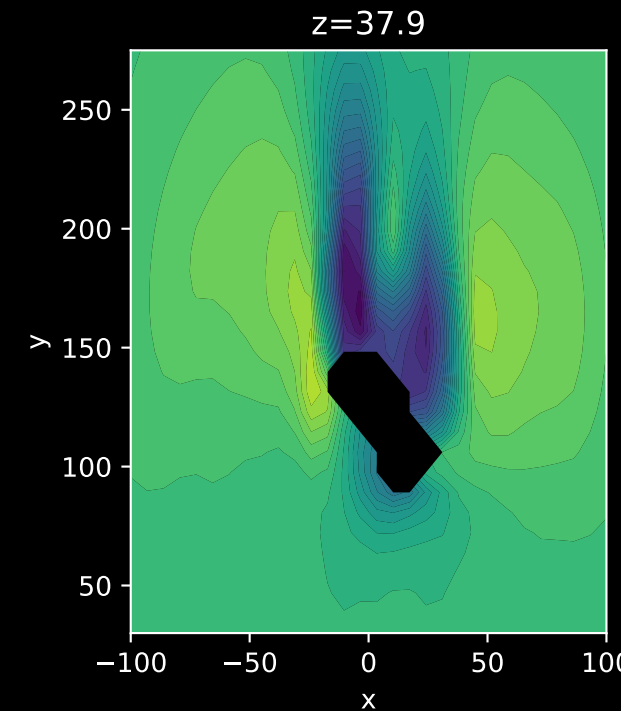
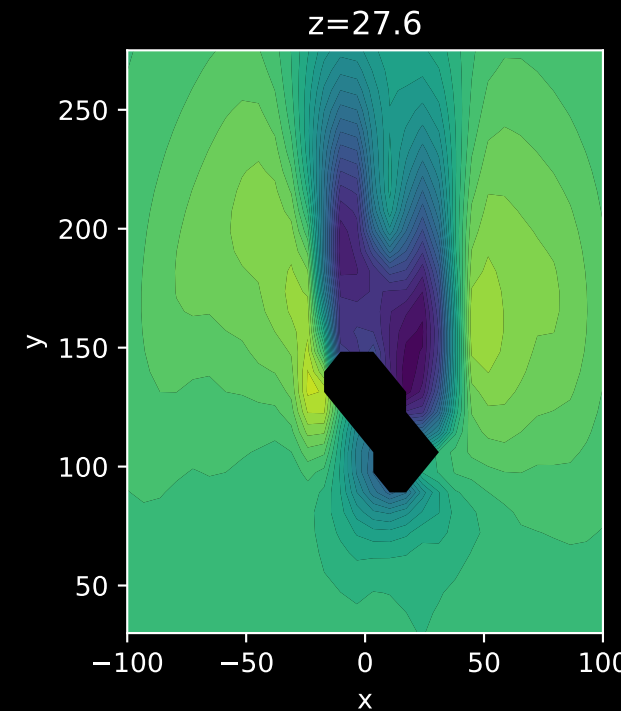
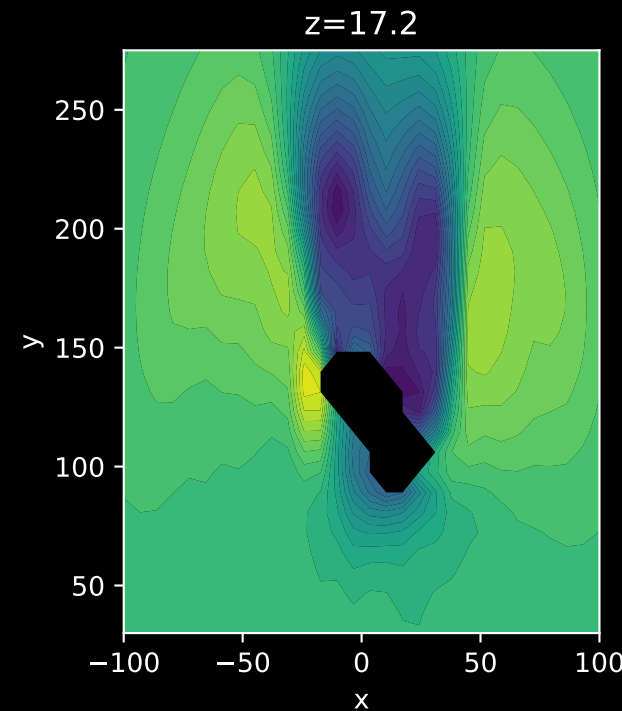
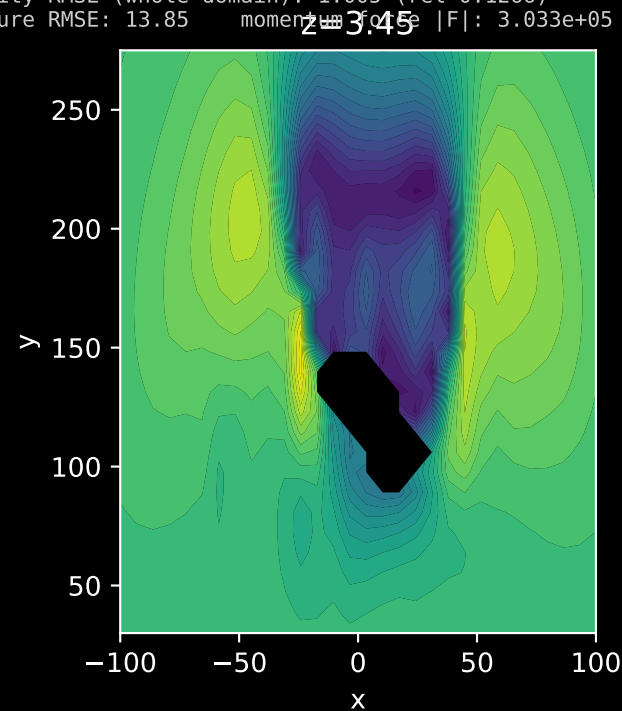


Posterior - CFD $|u|$, $z=24.14$



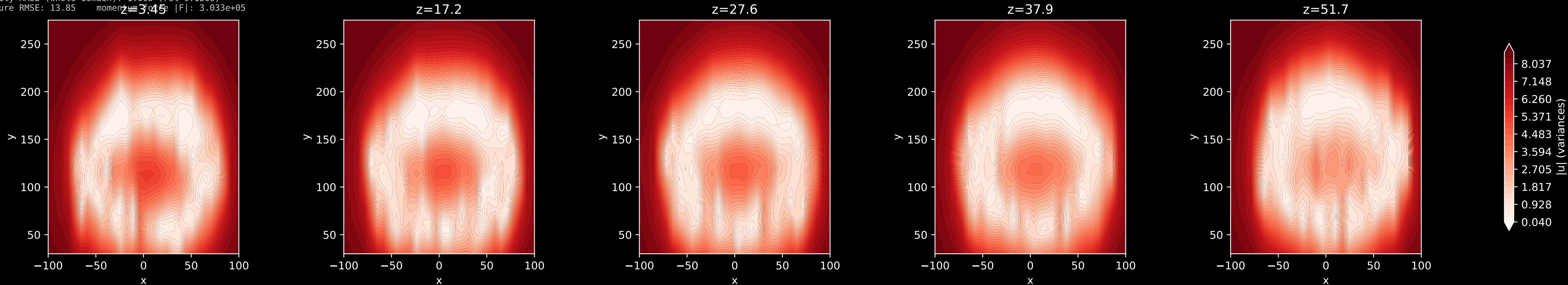
phase_topcap | 419 training pts
velocity RMSE (whole domain): 1.005 (rel 0.1266)
pressure RMSE: 13.85 moment_force |F|: 3.033e+05

posterior $|u|$ slices along z



phase_topcap | 419 training pts
velocity RMSE (whole domain): 1.005 (rel 0.1266)
pressure RMSE: 13.85 moment force |F|: 3.033e+05

variances $|u|$ slices along z



phase_topcap | 419 training pts
velocity RMSE (whole domain): 1.005 (rel 0.1266)
pressure RMSE: 13.85 momentum force |F|: 3.033e+05

